Sachin patel

Operation Research Scientist

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Profile summary

I am a Data Scientist with a strong academic foundation from IIT Bombay (M.S. in Operations Research) and a B.Sc in Mathematical Sciences (Operational Research) from University of Delhi. With over 1.5 years of experience, I have designed and deployed optimization and ML-driven solutions for logistics and scheduling problems, including VRP and MILP models using Pyomo and solvers such as Gurobi, CBC, and GLPK. I have built ETL pipelines with SQL and MongoDB, developed backend analytics APIs, and collaborated with cross-functional teams to deliver scalable decision-support systems. My strengths include optimization, time-series forecasting, machine learning, and production-ready data pipelines.

Skills

Python | Pyomo | Gurobi | CBC | GLPK | SQL | MongoDB | Machine Learning | Time Series Forecasting | Optimization | Pandas | NumPy | Scikit-learn | Seaborn | Matplotlib

Certifications

- Data Science for Engineers (NPTEL)
- Python for Data Science (IBM)
- Machine Learning training (Internshala)

Work Experience

Operation Research Scientist — Caliper Business Solutions **May 2023 - Jan 2025**

- Designed and implemented Vehicle Routing Problem (VRP) and Mixed-Integer Linear Programming (MILP) optimization models in Python using Pyomo

- Solved optimization models with Gurobi, CBC, and GLPK solvers to ensure efficient and scalable solutions

- Enhanced logistics planning efficiency by optimizing large-scale models, achieving up to 10% reduction in vehicle utilization and operational costs

- Developed and maintained backend APIs for GPS-based route analytics, encompassing data preprocessing, validation, and frequent-path identification

- Built robust ETL pipelines utilizing SQL and MongoDB to maintain high data quality for optimization and analytics workflows

- Integrated optimization and analytics components into production systems adhering to modular design, thorough testing, and comprehensive documentation standards

- Collaborated with cross-functional teams, including product managers and engineers, to deliver scalable decision-support solutions aligned with business goals

- Adhered to software engineering best practices by employing Git-based version control, modular code design, rigorous testing, and streamlined deployment workflows

Internships

Operations Research Intern - VRP solutions — Caliper Business Solutions **3 Months**

- Modeled and implemented customized Vehicle Routing Problem (VRP) solutions in Python using Pyomo framework, leveraging both commercial and open-source optimization solvers

- Collected, cleaned, and preprocessed data to enhance the accuracy and efficiency of optimization workflows

- Delivered solution strategies that achieved a reduction in vehicle usage by approximately 10%, improving operational efficiency and cost savings

Projects

Demand Forecasting for Inventory Optimization — 2 Months

- Created a demand forecasting system leveraging historical retail sales data (Walmart/M5) at product-store level.
- Improved forecast accuracy by **15-25%** over baselines using appropriate metrics (MAPE, SMAPE).
- Modeled trends, seasonality, and holidays for improved short- and long-term demand predictions.

Fraud Detection on Imbalanced Transaction Data — 6 Months

- Developed fraud detection models using Logistic Regression and tree-based classifiers for imbalanced transaction datasets.
- Achieved > **30% improvement in recall**, reducing costly false negatives.
- Used ROC-AUC and Precision-Recall metrics to align model selection with business risk.

Hospital Shift Scheduling Optimization — 10 Months

- Developed and implemented an MILP model to optimize multiday hospital staff scheduling.
- Achieved **10% cost reduction** compared to manual methods while ensuring coverage and adherence to constraints.
- Applied model to a 16-staff section and demonstrated scalability and adaptability to local labor rules.

Kernel Method for Canonical Correlation Analysis — 5 Months

- Proposed a kernel-based extension of CCA to capture nonlinear correlations, formulating it as an optimization problem.
- Conducted experiments showing improved performance over traditional CCA across datasets and compared kernel functions.

Optimization and Machine Learning Lab — 11 Months

- Designed and executed constrained optimization models using **Pyomo** and solvers (CBC, GLPK, IPOPT).
- Applied ML techniques including Linear/Logistic Regression, KNN, SVM, Random Forest, XGBoost, clustering, and PCA.
- Implemented optimization algorithms (gradient descent, Newton's, BFGS) from scratch for nonlinear problems.

Education

MS/M.Sc(Science) | Operation Research — **2024**
Indian Institute of Technology, Mumbai
Grade - 8.6/10

B.Sc - Bachelor of Science | Operational Research — **2022**
Deen Dayal Upadhyaya College, Delhi
Grade - 8.8/10

12th — **2019**
CBSE, English
Marks - 91%

10th — **2017**
CBSE, English
Marks - 84%

Extra-curricular activities

Department Fest Logistics

Managed food allocation, procured items, and assisted in certificate printing and guest arrangements for IEOR department fest.

Jawahar Navodaya Vidyalaya Admission

Cleared JNVST in 5th standard, securing admission and participating in school-level academic activities.

Achievements

Academic & Competitive Achievements

Secured AIR 60 in IIT JAM Mathematics 2022 and received Elite Silver in NPTEL Data Science for Engineers.

Additional information

Hobbies: Chess, Technical Blogging, Running

Languages: Hindi, English